

Quantum Field Theory – Homework 4

Name: _____ Due Date: 2025.11.17

The following problems are based on Lectures 7 and 8, focusing on the Hilbert space of the free scalar field and complex scalar field. The bullet points are guiding hints to help structure your answers.

Problem 1. Hamiltonian and Momentum Operators in Free Scalar Field Theory (20 points)

Consider a real free scalar field of mass m in $(3+1)$ dimensions with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

In the Heisenberg picture, the quantized field operator may be expanded as

$$\hat{\phi}(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}}t + i\vec{k}\cdot\vec{x}} + \hat{a}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}}t - i\vec{k}\cdot\vec{x}} \right),$$

where $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$.

(a) Derive the conjugate momentum field

$$\hat{\Pi}(x) = \frac{\partial\mathcal{L}}{\partial(\partial_0\phi(x))}.$$

Write down the equal-time canonical commutation relation between $\hat{\phi}(t, \vec{x})$ and $\hat{\Pi}(t, \vec{x}')$.

(b) Starting from the Hamiltonian density

$$\hat{\mathcal{H}} = \frac{1}{2}\hat{\Pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2,$$

construct the Hamiltonian operator

$$\hat{H} = \int d^3\vec{x} \hat{\mathcal{H}}.$$

Rewrite $\hat{H}$ in terms of the creation and annihilation operators $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$.

(c) The spatial momentum operator of the field is defined as

$$\hat{P}^i = \int d^3\vec{x} \hat{T}^{0i},$$

where the energy–momentum tensor for this theory is

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L}.$$

Using this expression, show that

$$\hat{T}^{0i}(t, \vec{x}) = \hat{\Pi}(t, \vec{x}) \partial^i \hat{\phi}(t, \vec{x}).$$

Then express $\hat{P}^i$ in terms of $\hat{a}_{\vec{k}}$ and $\hat{a}_{\vec{k}}^\dagger$.

(d) Define the one-particle momentum eigenstate

$$|\vec{k}\rangle = \hat{a}_{\vec{k}}^\dagger |0\rangle,$$

where $|0\rangle$ is the vacuum annihilated by all $\hat{a}_{\vec{k}}$. Show that $|\vec{k}\rangle$ is an eigenstate of both the Hamiltonian $\hat{H}$ and the momentum operator $\hat{P}^i$. Explain why $|\vec{k}\rangle$ is interpreted as a relativistic particle of mass m .

Problem 2. Quantization of the Complex Scalar Field (20 points)

Consider a complex scalar field in $(3 + 1)$ dimensions with Lagrangian density

$$\mathcal{L} = -\partial_\mu \phi \partial^\mu \phi^* - m^2 \phi^* \phi.$$

- (a) Treat ϕ and ϕ^* as independent classical fields. Derive the conjugate momenta

$$\Pi_\phi = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)}, \quad \Pi_{\phi^*} = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi^*)}.$$

After quantization, impose the equal-time canonical commutation relations

$$[\hat{\phi}(t, \vec{x}), \hat{\Pi}_\phi(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'), \quad [\hat{\phi}^\dagger(t, \vec{x}), \hat{\Pi}_{\phi^\dagger}(t, \vec{x}')] = i\delta^{(3)}(\vec{x} - \vec{x}'),$$

with all other equal-time commutators vanishing.

- (b) Solve the Klein–Gordon equation for the complex field. Write the most general operator expansion of $\hat{\phi}(x)$ as

$$\hat{\phi}(x) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(\hat{a}_{\vec{k}} e^{-i\omega_{\vec{k}} t + i\vec{k} \cdot \vec{x}} + \hat{b}_{\vec{k}}^\dagger e^{i\omega_{\vec{k}} t - i\vec{k} \cdot \vec{x}} \right),$$

and write the corresponding expression for $\hat{\phi}^\dagger(x)$.

- (c) Insert your mode expansions into the equal-time commutation relations from part (a). By comparing coefficients of independent plane waves, determine all non-vanishing commutators among the operators

$$\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger, \quad \hat{b}_{\vec{k}}, \hat{b}_{\vec{k}'}^\dagger.$$

Show that the result must be

$$[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad [\hat{b}_{\vec{k}}, \hat{b}_{\vec{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'),$$

with all other commutators equal to zero.